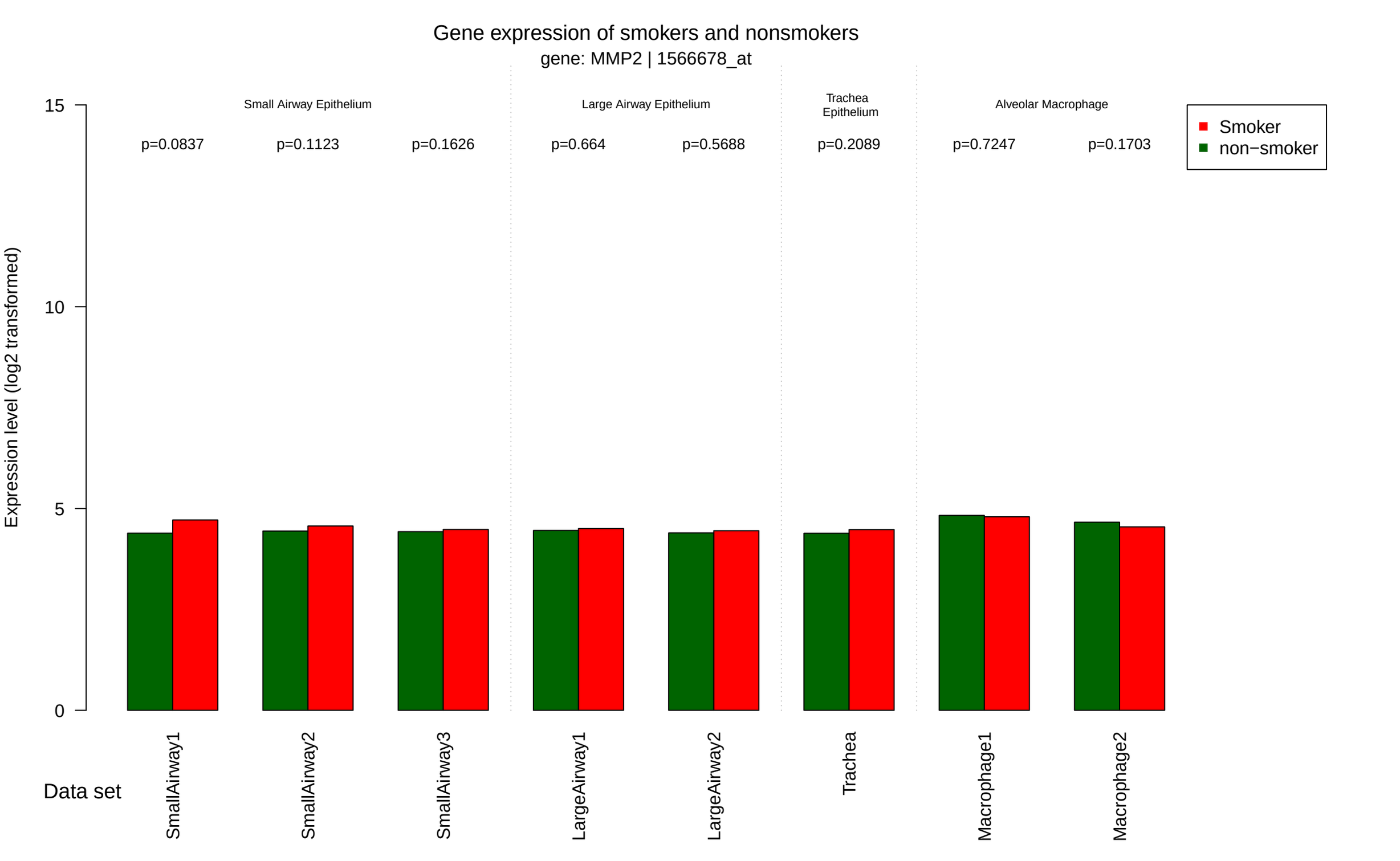
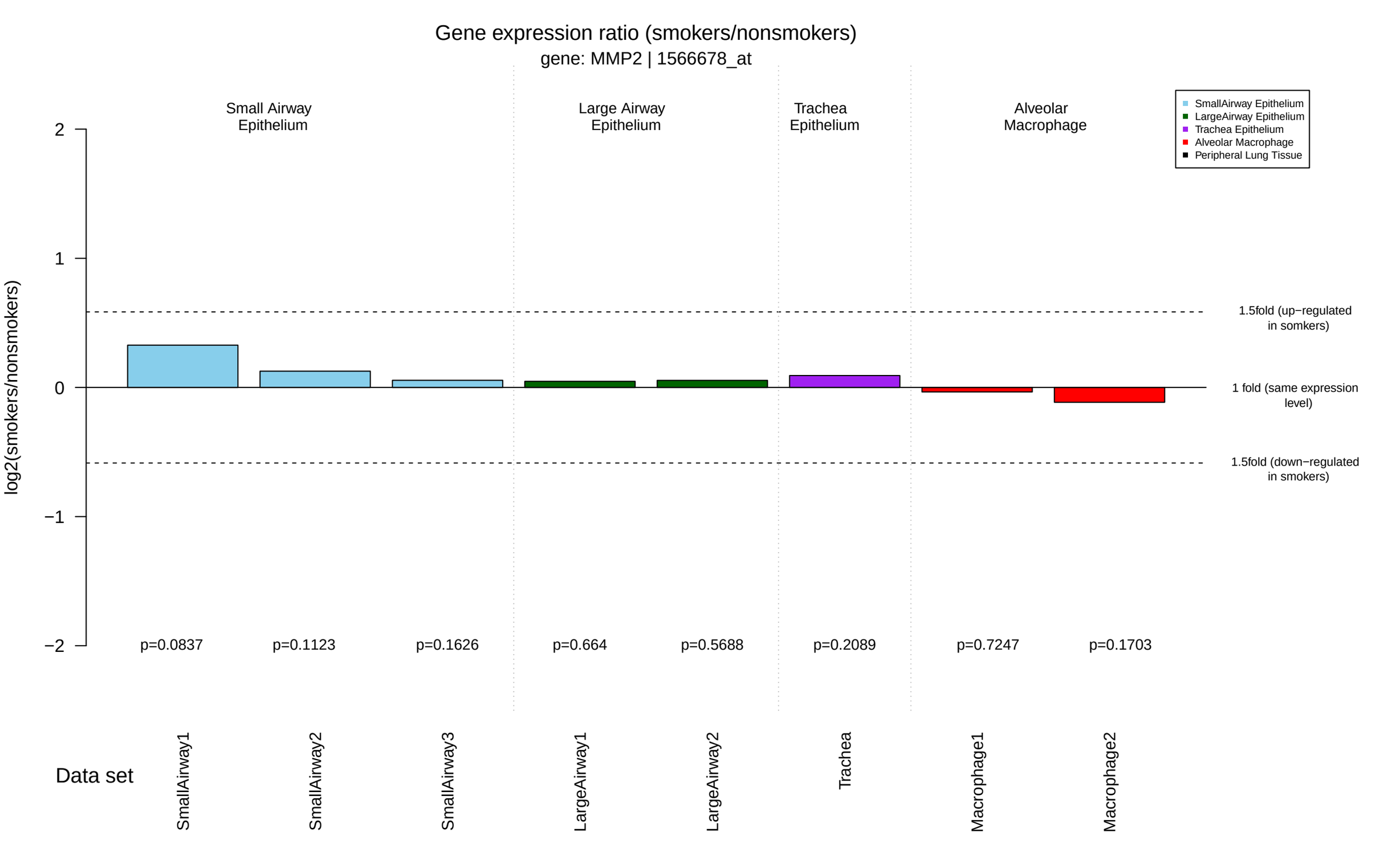
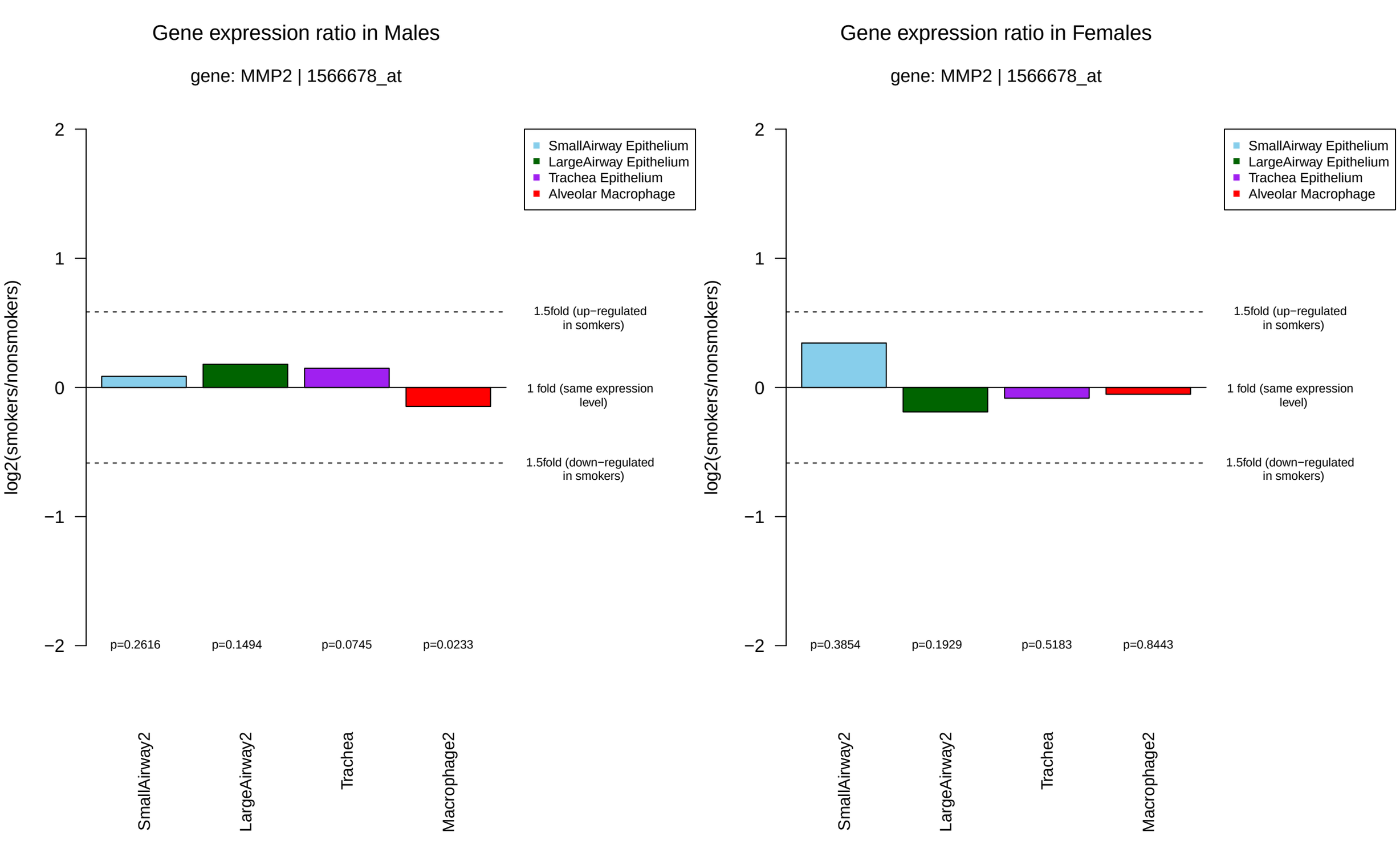
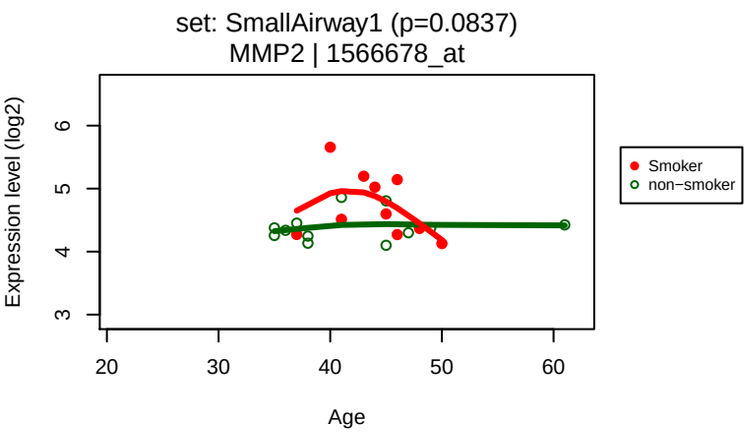
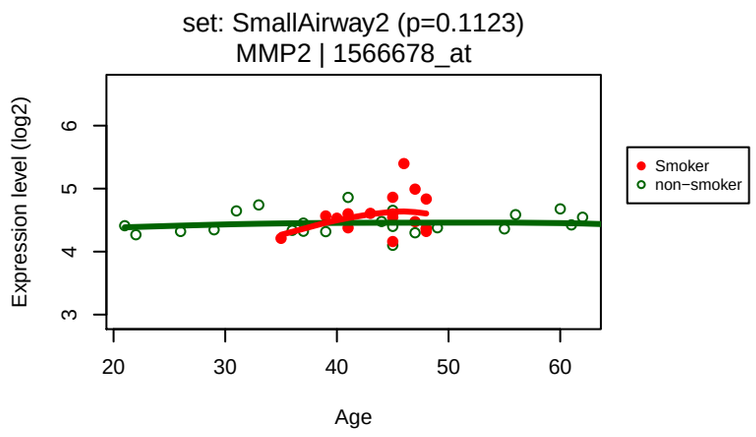
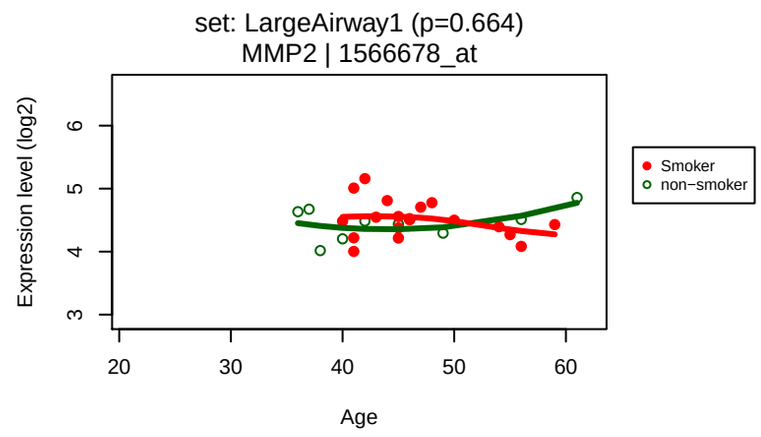

Gene expression of smokers and nonsmokers
gene: MMP2 | 1566678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0837
p=0.1123
p=0.1626
p=0.664
p=0.5688
p=0.2089
p=0.7247
p=0.1703
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

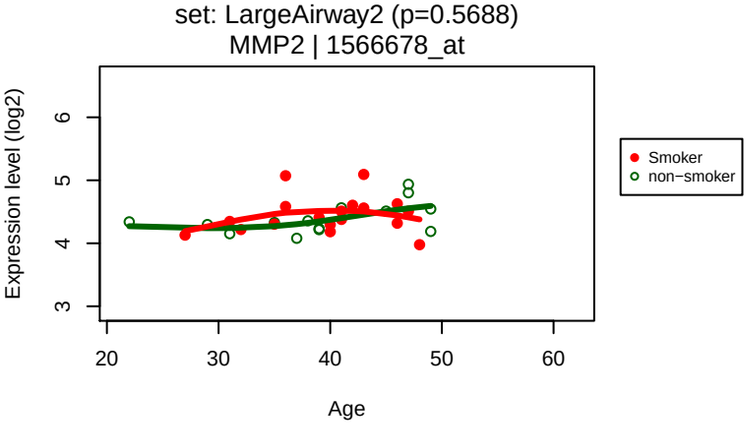

Gene expression ratio (smokers/nonsmokers)
gene: MMP2 | 1566678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0837
p=0.1123
p=0.1626
p=0.664
p=0.5688
p=0.2089
p=0.7247
p=0.1703
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

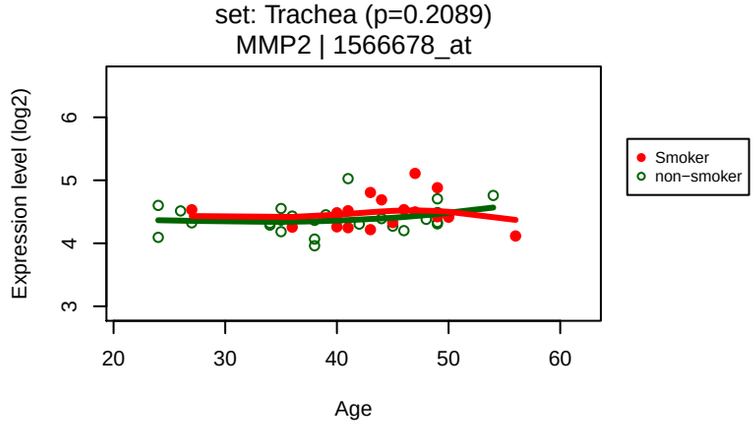

Gene expression ratio in Males
gene: MMP2 | 1566678_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2616
p=0.1494
p=0.0745
p=0.0233
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP2 | 1566678_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3854
p=0.1929
p=0.5183
p=0.8443
SmallAirway2
LargeAirway2
Trachea
Macrophage2

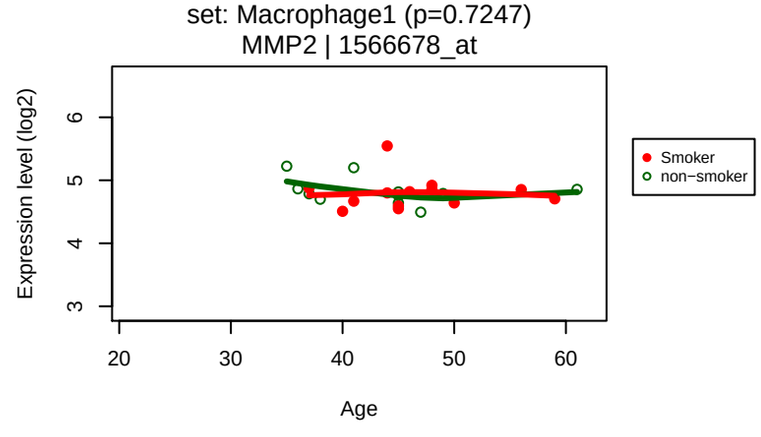

set: SmallAirway1 (p=0.0837)
MMP2 | 1566678_at
Expression level (log2)
Age
Smoker
non−smoker

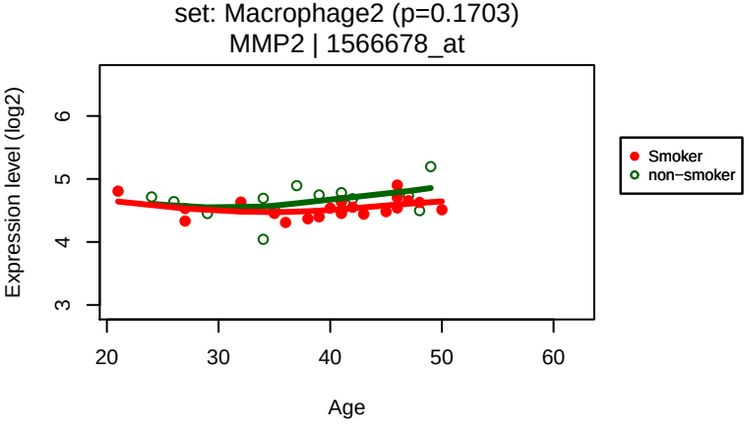

set: SmallAirway2 (p=0.1123)
MMP2 | 1566678_at
Expression level (log2)
Age
Smoker
non−smoker

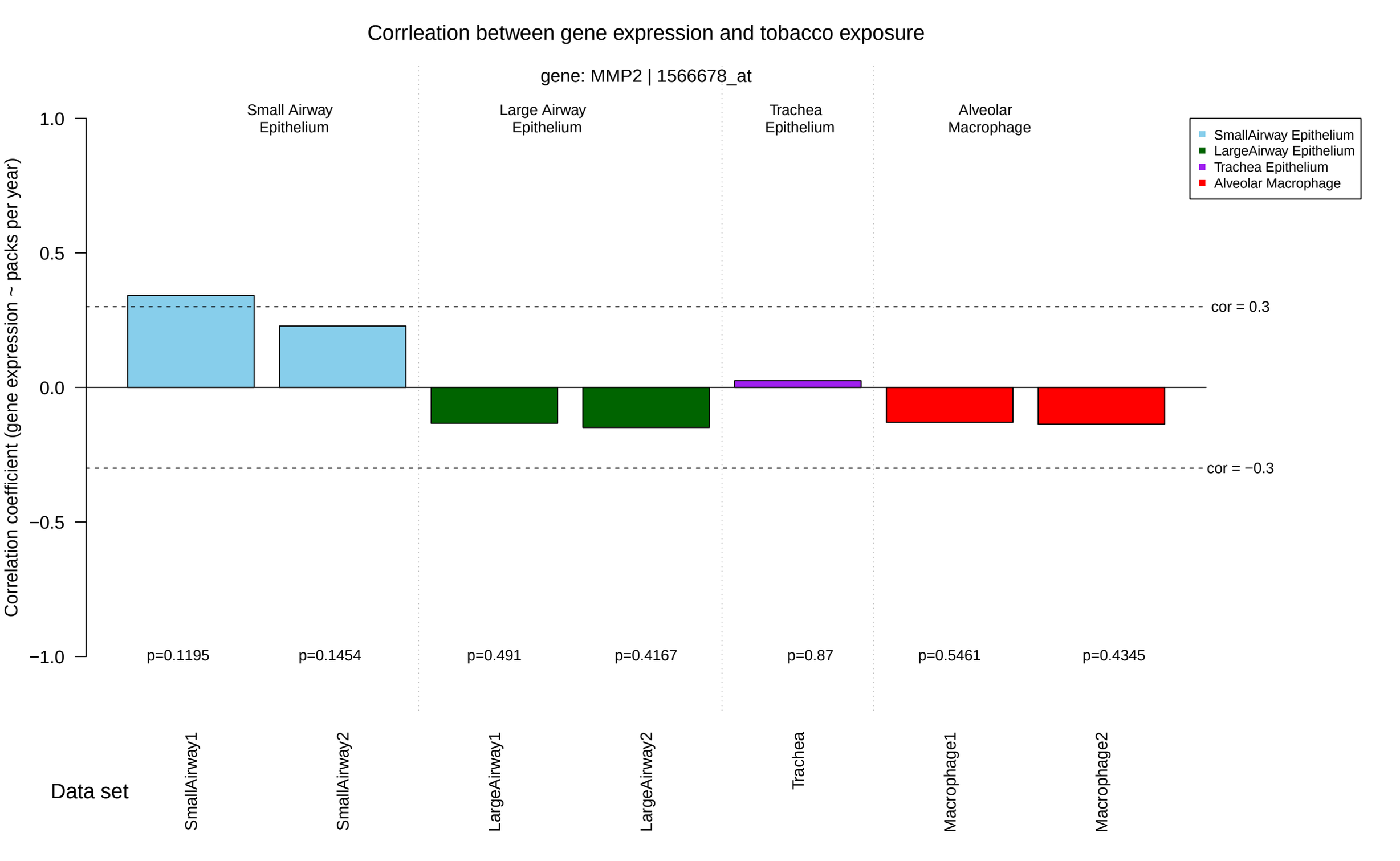

set: LargeAirway1 (p=0.664)
MMP2 | 1566678_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5688)
MMP2 | 1566678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2089)
MMP2 | 1566678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7247)
MMP2 | 1566678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1703)
MMP2 | 1566678_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP2 | 1566678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1195
p=0.1454
p=0.491
p=0.4167
p=0.87
p=0.5461
p=0.4345
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2